

Richard Rui Chu

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<https://scholar.google.com/citations?user=Jd2YfKQAAAAJ>

Education

Cornell University – Tufts University

Ph.D. in Computer Engineering (Advisor: Prof. Yingjie Lao) & Ph.D. in Systems Engineering
Dean's Fellowship; GPA: 3.9/4.0

Boston, MA, USA

2023 – 2026

Columbia University

M.S. in Computer Engineering

New York, NY, USA

2021 – 2023

University College London – Fudan University – University of Shanghai for Science and Technology

B.S. in Information Science; Dual Degree in Finance
London, UK; Shanghai, China

2016 – 2021

Research Interest

My research focuses on **Trustworthy and Efficient Foundation Models**: large language models (LLMs) and multimodal models that stay secure, accountable, and deployable once they leave the lab. I work across three layers of the deployment lifecycle: (1) exposing vulnerabilities, including adversarial hijacking of in-context learning and imperceptible backdoors in diffusion models; (2) building inference-time safeguards that need no retraining, including copyright-aware code generation, sparse watermarking for provenance, retrieval-augmented debiasing, and contextual unlearning; and (3) making models reliable and fast in physical and high-throughput settings, including attention steering for multimodal video understanding, neural evaluation of autonomous-driving planners, and FPGA–algorithm co-design for multi-agent LLM systems. These directions serve applications in **AI security, intellectual-property protection, fair and accountable AI, autonomous driving, and efficient AI infrastructure**. My research has been supported by the U.S. National Science Foundation (NSF).

Honors and Awards

- Highlight Paper, IEEE/CVF CVPR 2025, for *UIBDiffusion* (388 highlights out of 13,008 valid submissions, about 3%)
- Dean's Fellowship, Tufts University, 2023

Selected Publications

Note: In artificial intelligence and machine learning, peer-reviewed top-tier conferences are considered the premier venue of publication. NeurIPS, CVPR, ECCV, and EMNLP are flagship conferences listed at CSRankings.

- **Chu, R.**, et al. (2026, December). FASD: Hardware Acceleration for Multi-AI-Agent Discussion. In *Advances in Neural Information Processing Systems 39 (NeurIPS'26)*. To appear.
- **Chu, R.**, Meng, N., Tithy, T. D., Cao, T., Shezan, F. H., & Lao, Y. (2026). AegisCodeGen: Toward Copyright-Aware Code Generation with Runnability-Guided Fusion. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP'26)*. To appear.
- **Chu, R.**, & Lao, Y. (2026, September). HAS: Highlight-Guided Attention Steering for Multimodal LLM Video Summarization. In *Computer Vision – ECCV 2026, Lecture Notes in Computer Science* (pp. 138–157). Springer (*ECCV'26*). https://doi.org/10.1007/978-3-032-37041-9_8 (Acceptance rate: 2,883/10,473 = 27.5%)
- **Chu, R.**, Zhao, B., Jiang, H., Aeron, S., & Lao, Y. (2025, December). BAM-ICL: Causal Hijacking In-Context Learning with Budgeted Adversarial Manipulation. In *Advances in Neural Information Processing Systems 38* (pp. 15819–

15850) (*NeurIPS'25*). <https://doi.org/10.52202/085713-0476> (Acceptance rate: 5,290/21,575 = 24.5%)

- Han, Y., Zhao, B., **Chu, R.**, Luo, F., Sikdar, B., & Lao, Y. (2025, June). UIBDiffusion: Universal Imperceptible Backdoor Attack for Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 19186–19196) (*CVPR'25*). **Highlight Paper**. <https://doi.org/10.1109/CVPR52734.2025.01787> (Acceptance rate: 2,878/13,008 = 22.1%; Highlight: 388/13,008 = 3.0%)
- Hoang, D. C., Le, T. Q. H., **Chu, R.**, Li, P., Zhao, W., Lao, Y., & Doan, K. D. (2026, March). SpARK: An Embarrassingly Simple Sparse Watermarking in LLMs with Enhanced Text Quality. In *Findings of the Association for Computational Linguistics: EACL 2026* (pp. 4603–4626) (*EACL'26 Findings*). <https://doi.org/10.18653/v1/2026.findings-eacl.240>
- Wu, X., Kiden, S., **Chu, R.**, Williams, J., Zhang, S., & Da Fonseca, A. M. (2026, February). Explainability in Speech AI: Challenges and Future Directions. In *2026 IEEE International Conference on Semantic Computing* (pp. 320–327) (*ICSC'26*). <https://doi.org/10.1109/ICSC67292.2026.00053>
- Wu, Z., Wang, J., Qi, Q., Shu, M., **Chu, R.**, Li, J., Jin, J., & Chen, D. (2024, July). FlowRCA: Enhancing Microservice Reliability with Non-invasive Root Cause Analysis. In *2024 IEEE International Conference on Web Services* (pp. 1251–1258) (*ICWS'24*). <https://doi.org/10.1109/ICWS62655.2024.00149>
- Gao, Y., Ren, P., Shu, M., **Chu, R.**, Li, J., Jin, J., & Li, W. (2024, April). SO-Net: Model-Agnostic Sequential Hand Pose Optimization Framework. In *2024 IEEE International Conference on Acoustics, Speech and Signal Processing* (pp. 3755–3759) (*ICASSP'24*). <https://doi.org/10.1109/ICASSP48485.2024.10445741>
- Sun, J., Xing, Z., **Chu, R.**, Bai, H., Wang, J., & Peng, X. (2019, September). Know-How in Programming Tasks: From Textual Tutorials to Task-Oriented Knowledge Graph. In *2019 IEEE International Conference on Software Maintenance and Evolution* (pp. 257–268) (*ICSME'19*). <https://doi.org/10.1109/ICSME.2019.00039> (Acceptance rate: 31/135 = 23%)

Manuscripts Under Review

- **Chu, R.**, Zhao, B., Le, T. Q. H., Hoang, D. C., Lin, H., Li, P., Zhao, W., Doan, K. D., & Lao, Y. (2026). DebiasRAG: A Tuning-Free Path to Fair Generation in Large Language Models through Retrieval-Augmented Generation. Under review at *SIGMOD 2026*. arXiv:2605.16113. <https://arxiv.org/abs/2605.16113>
- **Chu, R.**, et al. (2026). D0CUR: Zero-Shot LLM Unlearning via Dynamic Contextual Retrieval. Under review at *AAAI 2027*.
- **Chu, R.**, et al. (2026). Multi-Agent System is a Few-Shot Dimensional Learner. Under review at *NAACL 2026*.

Professional Experience

Motional	<i>Boston, MA, USA</i>
Research Scientist in AI and Autonomous Driving	May 2026 – Aug 2026
Tesla	<i>Austin, TX, USA</i>
Data Streaming Engineer: DevOps and Model Deployment (University Collaboration Position)	Summer 2023
Virufy AI-Health @ Stanford University	<i>Silicon Valley, CA, USA</i>
Model Deployment Engineer	Summer 2022
Bosch	<i>Global</i>
Embedded Engineer	May 2021 – Sep 2021

Professional Service

Organizing Committee

- Co-Host, DCASE Conference (Detection and Classification of Acoustic Scenes and Events)

Program Committee

- PC Member, ECIR (European Conference on Information Retrieval)

Reviewer

- Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Machine Learning (ICML)
- International Conference on Learning Representations (ICLR)
- Conference on Empirical Methods in Natural Language Processing (EMNLP)
- ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)

Teaching

Tufts University*Medford, MA, USA***Teaching Assistant**

- EE 141: Trusted and Responsible AI Spring 2026
- EE 193: AI Security and Privacy Spring 2025

Columbia University*New York, NY, USA***Teaching Assistant**

- COMS 4735: Visual Interfaces to Computers (Prof. John Kender) Spring 2023
- EE 6889: Stream Processing Spring 2023
- EECS 6893: Big Data Analytics and Advanced AI Fall 2022

Research Experience

Tufts University, Department of Electrical and Computer Engineering

Medford, MA, USA

Ph.D. Researcher (advisor: Prof. Yingjie Lao)

2023 – Present

- **Budgeted Adversarial Hijacking of In-Context Learning (BAM-ICL)**: Identified that existing hijacking attacks on LLM in-context learning (ICL) spread a fixed perturbation budget across in-context examples (ICEs) and miss the gains of reallocating it. Designed a two-stage budgeted adversarial manipulation framework: an offline, global gradient-based attack learns the optimal budget profile across ICEs, and an online stage generates perturbations progressively as ICEs arrive one by one, a stricter and more realistic threat model in which only the current example can be modified. Across diverse LLMs and datasets, the attack achieved higher success rates and stealthiness while keeping text quality, and the adversarial ICEs transferred to other models. (**first author paper** at NeurIPS'25)
- **Highlight-Guided Attention Steering for Video Summarization (HAS)**: Proposed an attention-steering method for multimodal LLMs (MLLMs) that replaces discrete key-frame selection with a continuous, frame-level highlight distribution computed over the whole video. Injected this distribution at inference time as an attention steering vector, so the MLLM attends more to highlighted frames while still retaining information from the rest of the video. Evaluated on a range of video-summarization benchmarks. (**first author paper** at ECCV'26)
- **Copyright-Aware Code Generation (AegisCodeGen)**: Formulated copyright protection for code LLMs as a region-aware generation-control problem: first locate where code can be safely modified, then decide how strongly to apply copyright-aware fusion. Built an inference-time framework that samples multiple generations to construct an agreement profile over the target program, then uses that profile to gate logit-fusion strength, with stronger protection in low-agreement regions and weaker shifts near executable anchors. Reduced reproduction of protected code while preserving runnability and generation utility, with no retraining and no access to training data. Collaborated with researchers at UT Arlington and the Tufts Fletcher School of Law and Diplomacy. (**first author paper** at EMNLP'26)
- **Hardware Acceleration for Multi-Agent LLM Discussion (FASD)**: Developed an FPGA–algorithm co-design framework that cuts the latency of discussion in LLM-powered multi-agent systems (MAS). Modeled MAS discussion as a graph-coupled dynamical system so that agent-activation pruning can be optimized through phase-coupled dynamics, profiled representative MAS workflows to build phase profiles, and mapped the pruning operator onto an FPGA-friendly datapath. Improved discussion efficiency while preserving task performance, with larger gains as the agent graph grows in size and complexity. (**first author paper** at NeurIPS'26)
- **Imperceptible Backdoor Attacks on Diffusion Models (UIBDiffusion)**: Contributed to a universal imperceptible backdoor attack for diffusion models that turns universal adversarial perturbations (UAPs) into image- and model-agnostic triggers. A single trigger works across diffusion models and samplers, matches prior attacks in generation quality (FID) with higher attack success at low poison rates, and evades the state-of-the-art defenses Elijah and TERD. (**co-authored paper** at CVPR'25, **Highlight**)
- **Provenance, Fairness, and Unlearning for LLMs (SpARK, DebiasRAG, D0CUR)**: Co-developed SpARK, a sparse LLM watermark that anchors watermarked tokens to words with specific part-of-speech tags, reaching high detectability with better text quality than prior watermarking schemes. Led DebiasRAG, a tuning-free, query-specific debiasing framework built on retrieval-augmented generation: it derives debiasing contexts from self-diagnosed bias contexts and reranks retrieved context with gradient-updated debiasing guidance, reducing social bias (race, gender, age) while preserving the LLM's representation ability. Led D0CUR, a zero-shot LLM unlearning method based on dynamic contextual retrieval. (**co-authored paper** at EACL'26 Findings; **two first author manuscripts** under review)

Motional

Boston, MA, USA

Research Scientist in AI and Autonomous Driving

May 2026 – Aug 2026

- **Neural Evaluation for Autonomous-Driving Planning (MotionScorer)**: Designed an end-to-end neural-network evaluation system as a component of Motional's driving world model. Integrated the evaluator into a vision-language-model (VLM) semantic workflow and distilled it into a scorer for the autonomous vehicle (AV) planner. Evaluated human-aligned trajectory scoring on public benchmarks, including the Waymo Open Dataset end-to-end driving (WOD-E2E) data, and analyzed behavior across semantic-disagreement regimes and under causal readout interventions. (**manuscript** in submission)

- Tesla** *Austin, TX, USA*
Data Streaming Engineer: DevOps and Model Deployment Summer 2023
 - University collaboration position. Established cloud-computing data pipelines for manufacturing software, and built model-deployment and data-streaming pipelines that connect trained models to production systems.
- Virufy AI-Health @ Stanford University** *Silicon Valley, CA, USA*
Model Deployment Engineer Summer 2022
 - Trained a cough-audio diagnostic model in TensorFlow, deployed it on edge devices, and built the supporting AWS web services. Collected more than 5,000 COVID-19 cough recordings and labeled them for training; tracked and scheduled work in JIRA.
- Bosch** *Global*
Embedded Engineer May 2021 – Sep 2021
 - Improved the performance of Bosch power-tool systems by 5% using data collected from hardware, and tested hardware functions.
- Claudius Legal Intelligence AI, NLP Lab @ Princeton University** *Boston, MA, USA*
Research Assistant (with Prof. Melanie Weber, Harvard University)
 - Retrained legal-domain language models in the cloud on legal papers and materials, building on prior models.
- Creative Machines Laboratory, Columbia University** *New York, NY, USA*
Researcher (with Prof. Hod Lipson & Dr. Phillipe)
 - Parallelized robot simulation environments in Isaac Gym, using CUDA and OpenGL, to accelerate reinforcement-learning training of robots.
- Columbia University** *New York, NY, USA*
Research Assistant (with Prof. Sharon Di)
 - *Reinforcement Learning for Autonomous-Driving Routing in Simulation*: Built a Unity simulation environment to train a car to avoid obstacles and applied a Transformer model. Implemented DQN and DDPG in PyTorch with CUDA and improved the models through fine-tuning.
- Columbia University & Northwestern University** *New York, NY, USA*
Research Assistant (with Prof. Gil Zussman & Prof. Igor Kadota)
 - *Computer Vision for Network-Fluctuation-Based Wind Prediction*: Trained ImageNet-based computer-vision models to classify and predict wind maps generated from mmWave network measurements, and applied reinforcement learning to MIMO beamforming at base stations.
- University College London** *London, UK*
Project Leader (with Google and Prof. Josep Grau-Bove)
 - *Virtual Reality Design for the British Museum*: Designed an improved 3D virtual-reality tour of the museum using Unity and 3D sensor fusion.